

Sem 2 AY23-24 MH1101 Calculus II Test (1 hour)

Name:

Matriculation Number:

Tutorial Group:

Question 1 [16 marks]

(a) Write the following limit as a definite integral (do not evaluate it).

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{8}{\sqrt{9n^2 - 4i^2}}$$

(b) If $f(x) = (\sin x) \cdot \int_{x^2}^1 \sqrt{1 + \cos(4t)} dt$, find $f'(\frac{\pi}{2})$. You may leave your final answer in terms of cosines.

Answer.

More space for answers.

Question 2 [16 marks]

- (a) Let R be the region in the xy -plane bounded by the curves $x = 3y^2 - 2$ and $x = y^2$. Compute the volume of the solid formed by revolving R about the horizontal line $y = 1$.
- (b) Determine if the following improper integral converges, and if so, evaluate it. Justify your answer. (You may consider substitution $u = e^x$.)

$$\int_{-\infty}^{\infty} \frac{1}{e^x + e^{-x}} dx.$$

Answer.

More space for answers.

More space for answers.

Question 3 [18 marks] Evaluate the following integrals. Lay out the steps clearly and justify your answers.

(a) $\int_0^{\pi/4} \tan^5(x) dx$

(b) $\int \sin(\ln x) dx$ (Express your final answer in terms of x .)

(c) $\int_0^1 x^m (\ln x)^n dx$, where m, n are positive integers. Express your final answer in terms of m and n .

Answer.

More space for answers.

More space for answers.

More space for answers.

Formulae Table

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int k dx = kx + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \tan x dx = \ln |\sec x| + C$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1} x$$

$$\int e^x dx = e^x + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \cos x dx = \sin x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \cot^2 x = \csc^2 x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\sin 2x = 2 \sin x \cos x$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\sin A \cos B = \frac{1}{2}(\sin(A - B) + \sin(A + B))$$

$$\sin A \sin B = \frac{1}{2}(\cos(A - B) - \cos(A + B))$$

$$\cos A \cos B = \frac{1}{2}(\cos(A - B) + \cos(A + B))$$

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

– End of Test –